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# Ultrasonic Projection for the Blind & Visually Impaired: A Proposed Model (Blind-Bat)

Yasser El-Sayed El-Alfy
Ahmed Abdulla Al-Goud

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# Ultrasonic Projection for the Blind & Visually Impaired: A Proposed Model (Blind-Bat)

By:

Yasser El-Sayed El-Alfy, 201720055

Ahmed Abdulla Al-Goud, 201810348

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# Approval Page

This project report has been submitted to Ahlia University by:

Yasser El-Sayed El-Alfy & Ahmed Abdulla Al-Goud

As a partial fulfillment for the degree of Bachelor in Mobile & Network Engineering and the degree of Bachelor in Computer & Communication Engineering and has been examined and approved by the examining committee:

Supervisor: Dr. Ammar Aldallal

Internal Examiner #1 Dr. Ahmed Jameel

Internal Examiner #2 Dr. Ayman Alaiwi

Dr. Alaeddine KAMARBU

Date: 26/12/2020

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# Abstract

Blind and visually impaired people face more difficulties and problems compared to others in society. Therefore, we are proposing a system that can help them live their lives normally like other people. Therefore, our aim is to design a system capable of assisting the blind and visually impaired people in navigating their surroundings. Furthermore, to achieve our aim we started by collecting the components that are suitable for our system, components such as Arduino, ultrasonic sensors, DC motors and vibrators. The Arduino will conduct the hardware and coding part and will be programmed and linked to the ultrasonic sensor and sends feedbacks to the vibrators. Finally, after testing the system, the result that is obtained was promising. We placed an object in front of the user and the user felt the vibration as an alert that there was an obstacle ahead. In conclusion, this system can be used to help the visually impaired person to have a life quality equally to any other person, which can be the only fair thing to do.

# 1 Chapter One: Introduction

## 1.1 Introduction

When a society is faced with a problem, it usually gets an attention especially if it affects a big number of people. So people would get together to try to solve their problem as quickly as possible. Disabled people face more problems in their lives than other people in society and they are helpless on their own. Interestingly, during our research we found out that there are various technologies that can be used to assist blind people. Some articles mentioned the use of haptic feedback which is very useful for the visually impaired person. As Fredrick Lenz said, “There is a world of deaf, dumb and blind people out there, who are shuffling through their lives, not knowing how wonderful life can be. They have their moments, but they are very unconscious”. According to World Health Organization (WHO), “Globally, at least 2.2 billion people have a vision impairment or blindness, of whom at least 1 billion have a vision impairment that could have been prevented or has yet to be addressed”. Therefore, in this project we will try to shed some light on different approaches and systems designed to help the blind or the visually impaired person who is unable to see what is around him/her and have difficulties moving around without any help. To that end, some new and advanced technologies were developed to help visually impaired people. One of the approaches used in this field was developing a Metric system. Metric systems work by measuring distances and angles then giving a feedback on whether or not there is an obstacle in front of the person. Therefore, the metric systems can help the visually impaired person make some limited decisions based on his surroundings with the help of a sound or vibration that will alert him if he is close to an obstacle. Another approach which can be used is the Biotechnology. Biotech uses biological systems, living organisms to manufacture products or technologies which can improve the quality of human life. An example of a biotech is using a device that replaces the eye (i.e. a camera) and feed images directly into the optic nerves. Our aim is to find a low-cost, discreet, easy solution to our problem and also try to find whether there have been any similar proposals done for this problem. Brito, et al. (2018), have tested their first prototype application which allowed having some feedback from their users after trying their prototype. Results of the tests were a

success from which the feedback was gathered from usage of the application which helps visually impaired people in transportations and public places.

## 1.2 Problem Statement

Blind and visually impaired people are unable to view their surroundings, which makes it difficult to move around and navigate their surroundings without any assistance (i.e. human or technology).

Most of the researches done in this area take one of two approaches:

1. Biotechnology: replace the eye with a piece of hardware (i.e. a camera) that takes images and processes them then feeds some output directly into the optic nerves.
2. Metrics: measuring distances and angles then giving an output as a single binary feedback (i.e. can walk in this direction or not).

A worthy point for inquiry: how to project a live Image of the surroundings for blind and visually impaired people with a multi-binary feedback system using the metric approach.

## 1.3 Objectives

We are trying to develop a solution for blind people, which are aimed to ease their lives and help them navigate independently. It also aims to provide the blind person with a new experience and a way to view the surrounding world. So, we can sum our objectives in the following points:

- To design a system capable of assisting the blind and visually impaired people in navigating their surroundings.
- To develop a system that gives the blind people a close experience of the sense of sight by capturing the surrounding area then projecting in on the blind person's body which provides him with a bat-like sense of the area that he is moving in and trying to navigate.
- To investigate if such a system will be capable of smoothing and easing blind people's lives, by making the daily tasks done by other people such as walking around easy and comfortable for blind people.

## 1.4 Methodology

To conduct a hardware and a code associated with it and test it on an individual. This will be achieved through using Arduino programming, two ultrasonic sensors, a servo motor, a battery and some feedback vibrators which will all be securely positioned on a vest or a jacket that the blind person will be wearing while moving around and navigating his way throw the obstacles. This system has a big potential to help blind people if implemented in real life.

## 1.5 Limitations

Due to the pandemic of covid-19, we had some limitations in our project. First of many, as partners in the project, we had a problem in working together because one of us lives in Bahrain and the other lives in Saudi Arabia. In addition, we were aiming to work with ultrasonic speakers to produce haptic feedback but we could not get them because either we cannot ship them or they were overpriced. Therefore, we settled on using vibrators instead.

## 1.6 Project Timeline

*Table 1-1: Timeline.*

| Task name | September | October | November | December |
|---|---|---|---|---|
| • **Searching and buying appropriate equipment with low cost** | | | | |
| • **Testing, fixing, and enhancing the project** | | | | |
| • **Finish implementing the project** | | | | |
| • **Writing the report** | | | | |
| • **Presentation of project** | | | | |

## 1.7 Report Structure

Chapter 2 (LITERATURE REVIEW): Stating the previous findings and researches that was done in the field and the latest technology that has been implemented in real life.

Chapter 3 (System Components): Listing all the components that has been used in the project and the overall cost.

Chapter 4 (System Implementation): How the mentioned components in chapter 3 has been used in association with the software to build the system.

Chapter 5 (Conclusion): Stating the findings and the obtained results with the future work that can improve the system.

# 2 Chapter Two: Literature Review

## 2.1 Introduction

This chapter covers our method of building the system (Blind Bat), plus sources on different methods of how to have a system that can help the disabled and the visually impaired to have a better life.

## 2.2 Blind Bat

The blind bat system can be used by everyone that is disabled and visually impaired. We chose to do this system to help improve their lives. The blind bat system has sensors and vibrators that must be worn to detect any obstacles close by and give a feedback.

The main components of the system are vibrators, ultrasonic ranging module, servo motor and an Arduino.

This system is just a prototype and there is still much work and effort needed in this project to make it more efficient.

## 2.3 Literature Review

Different researches and systems have been developed in this field. An overview will help determine whether our solution to this problem is helpful or not.

According to Hoang, et al. (2016), who developed an assistive system that helps the blind and visually impaired in detecting obstacles, said that the system has been tested and all of their 20 volunteers interacted well with the device. In addition, not just static obstacles are detected, some systems can even detect moving object. A computer vision-based system navigates for the visually impaired people and detects obstacles and moving objects. This system has been tested in outdoor urban places by volunteers. It was a challenging test because it captures multiple static and moving objects, but system features a real time detection which helps the users. (Tap, et al. 2009).

Furthermore, according to Habib, et al. (2019), they were able to develop a system that detects staircases and ground for blind people which is very useful because blind person will get notified by knowing the level of ground in front of him.

Moreover, Petsiuk and Pearce (2019) used a cheap and open-source system that helped the visually impaired navigates his surroundings. This system is a wearable ultrasound-based navigational system, it measures distances and converts it to haptic feedback with various patterns of vibration within four meter limit. This system has been tested by blindfolded volunteers that participated in different everyday tasks and were done successfully. This system is affordable because the components that are been used are cheap which can help many visually impaired people and can be used by a blind and deaf person.

However, according to Norberg, et al.(2016), using the haptic feedback the vibration rates are a bit slow but still possible to use by the visually impaired person. Furthermore, another way to help solve our problem is using a smart cane. According to, Akash & Rahman (2018), the smart cane is able to communicate with the visually impaired person with either sound or vibrations. The cane has been tested and it detects obstacles up to two meters efficiently.

In addition, according to Ramadhan (2018), a wearable smart system can be developed to assist the virtually impaired people. This system consists of various sensors, GPS & cellular communication. The sensors can track the user and alert them if any obstacle is around them by vibrations or sound. Moreover, a survey is been done between wearable and portable assistive devices for the visually impaired people. According to the results, it shows that none of the systems were 100% pleasingly in terms of efficiency but the device with more fundamental features will have a great performance (Elmannai, Elleithy 2017).

Furthermore, studies have analyzed different vibration patterns to guide blind people. They have placed different vibration patterns on the back and waist of the blind user to test which way it will be helpful to guide the user (Dura-Gil, et al. 2017).

This shows that there are many ways to help the blind people and most of the solutions give us different ideas but have the same output, which is helping the visually impaired. Therefore, with these many ideas, at the end of the day the users can choose

which way is most suited for them whether they are looking for the most comfortable system or they can go with what they can afford which is still a good path.

# 3 Chapter Three: System Components

## 3.1 Introduction

This chapter will list all the components that were utilized in order to build the prototype system. It includes the system components in addition with the description and specification of each of them.

## 3.2 Required Parts

Each component we used is important in order to build our project as follows:

### 3.2.1 Arduino Uno

*Figure 3-1: Arduino Uno.*

## Overview:

Arduino Uno (Figure 3-1) is a microcontroller board based on the ATmega328P (datasheet). It has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a 16 MHz ceramic resonator (CSTCE16M0V53-R0), a USB connection, a power jack, an ICSP header and a reset button. It contains everything needed to support the microcontroller; simply connect it to a computer with a USB cable or power it with an

AC-to-DC adapter or battery to get started. You can tinker with your Uno without worrying too much about doing something wrong, worst case scenario you can replace the chip for a few dollars and start over again.

## Features:

*Table 3-1: Arduino Uno features.*

| | |
|---|---|
| Microcontroller | ATmega328P |
| Operating Voltage | 5V |
| Input Voltage (recommended) | 7-12V |
| Input Voltage (limit) | 6-20V |
| Digital I/O Pins | 14 (of which 6 provide PWM output) |
| PWM Digital I/O Pins | 6 |
| Analog Input Pins | 6 |
| DC Current per I/O Pin | 20 mA |
| DC Current for 3.3V Pin | 50 mA |
| Flash Memory | 32 KB (ATmega328P) of which 0.5 KB used by bootloader |
| SRAM | 2 KB (ATmega328P) |
| EEPROM | 1 KB (ATmega328P) |
| Clock Speed | 16 MHz |
| LED_BUILTIN | 13 |
| Length | 68.6 mm |
| Width | 53.4 mm |
| Weight | 25 g |

### 3.2.2 Servo Motor SG90

*Figure 3-2: Servo Motor SG90.*

## Overview:

Micro Servo Motor SG90 (Figure 3-2) is a tiny and lightweight server motor with high output power. Servo can rotate approximately 180 degrees (90 in each direction), and works just like the standard kinds but smaller. You can use any servo code, hardware or library to control these servos. Good for beginners who want to make stuff move without building a motor controller with feedback &gearbox, especially since it will fit in small places. It comes with 3 different horns (arms) and hardware.

## Features:

*Table 3-2: Servo Motor SG90 features.*

| | |
|---|---|
| Weight | 9g |
| Dimension | 22.2 x 11.8 x 31 mm approximately |
| Stall torque | 1.8 kgf·cm |
| Operating speed | 0.1 s/60 degree |
| Operating voltage | 4.8 V (~5V) |
| Dead band width | 10 µs |
| Temperature range | 0 ºC – 55 ºC |

### 3.2.3 Ultrasonic Ranging Module (HC-SR04)

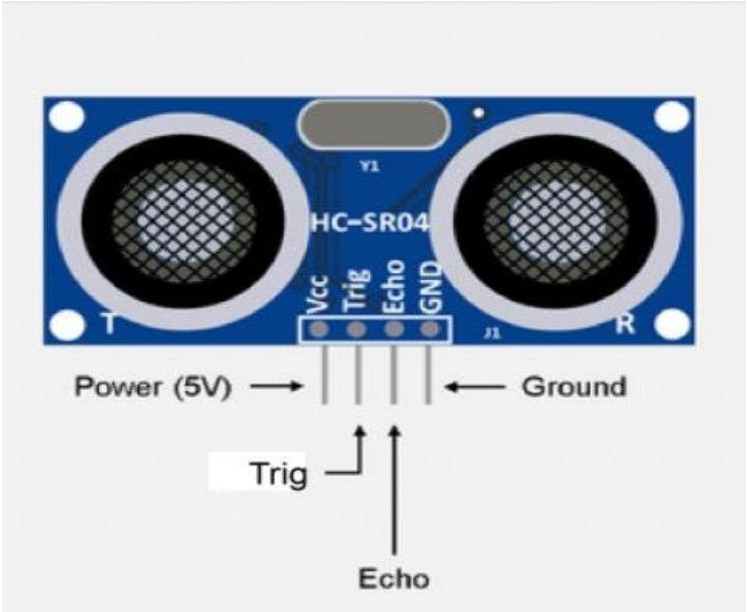


*Figure 3-3: Ultrasonic Ranging Module (HC-SR04).*

The Ultrasonic Ranging Module (Figure 3-3) provides stable readings and accurately measures distance. This module can be used for a number of projects such as fluid level in a tank, distance to obstacles, and other sensing tasks. Figure 3-4 show how the ultrasonic ranging module work, as you can see it sends out the signal and then bounces of an object and goes to the receiver and measures the distance by time.

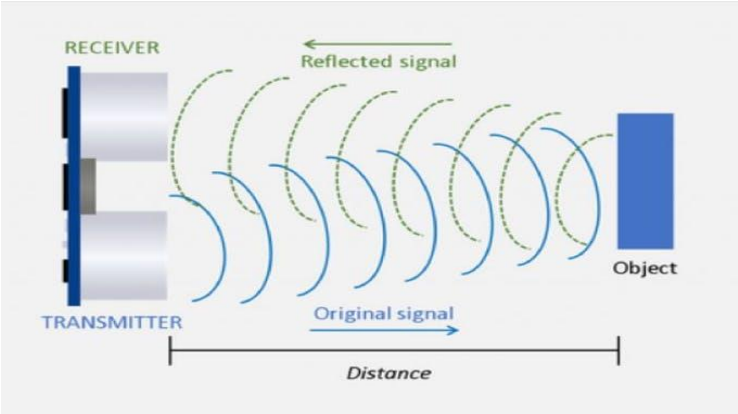


*Figure 3-4: Ultrasonic Ranging method of operation.*

## Features:

*Table 3-3: Ultrasonic Ranging features.*

| | |
|---|---|
| Working voltage | 5V(DC) |
| Static current | Less than 2mA |
| Output signal | Electric frequency signal, high level 5V, low level 0V |
| Sensor angle | Not more than 15 degrees |
| High precision | Up to 3mm |
| Mode of connection | VCC / trig(T) / echo(R) / GND |

### 3.2.4 Mini Vibration Motors

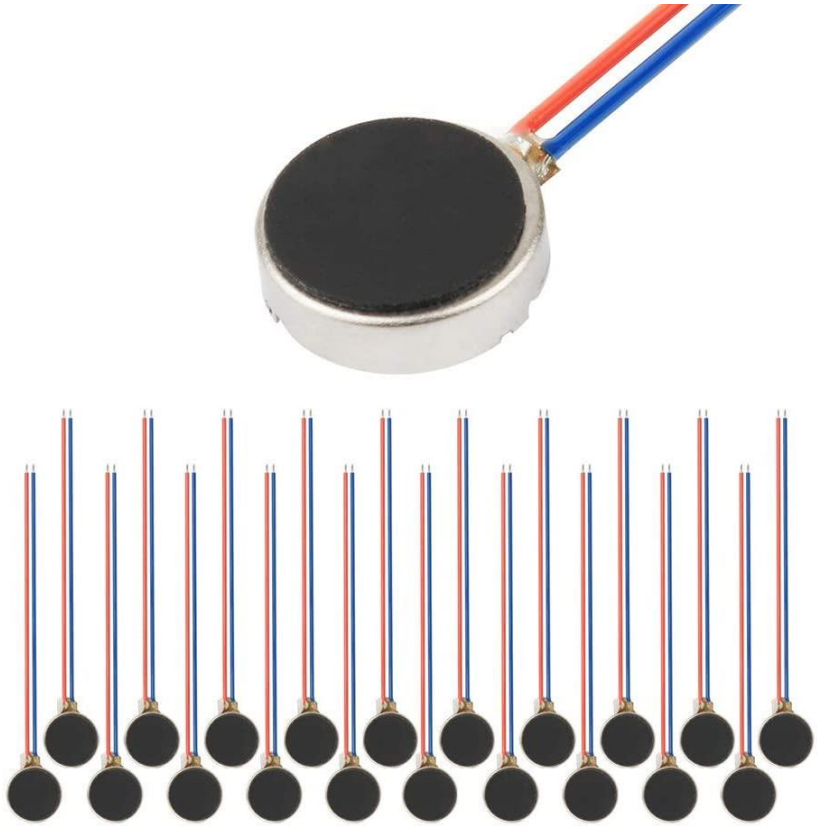

*Figure 3-5: Mini Vibration Motors.*

## Overview:

This mini vibration motor (Figure 3-5) worked at 3V-5V and can be used for multipurpose. As you can see in Figure 3-6, it shows in detail the components of the mini vibration motors.

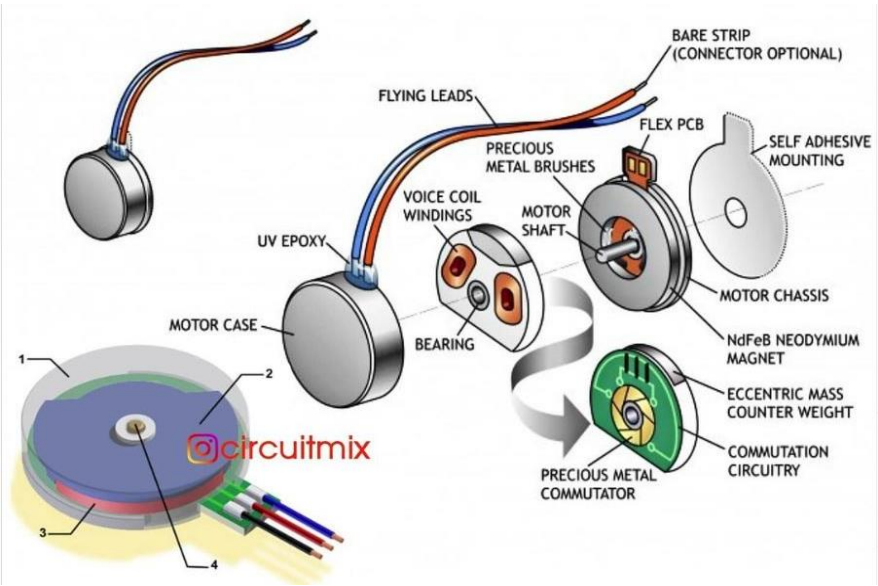


*Figure 3-6: Mini Vibration Motors components.*

## Features:

*Table 3-4: Mini Vibration Motors features.*

| | |
|---|---|
| Micro Vibration Motor Rated voltage | 3V; Rated Speed: 12000 RPM |
| Body Size | 10mm x 3mm/ 0.39" x 0.12" (D*T);Line length; 5 cm |
| Voltage – Rated | 1.3VDC ~ 3VDC |
| Package included | 20 x Cell Phone Vibration Motors |

### 3.2.5 PowerCore 10000mAh Power Bank

*Figure 3-7: PowerCore 10000mAh Power Bank.*

## Features:

*Table 3-5: PowerCore 10000mAh Power Bank features.*

| | |
|---|---|
| Product Dimensions | 9.2 x 6 x 2.2 cm ; 181 g |
| Boxed-product Weight | 200 g |
| Batteries | 1 Lithium ion batteries required. (included) |
| Item model number | AK-A1263011 |

## 3.3 Prototype Demonstration

The prototype system was demonstrated by using mini vibrators attached to a vest to help the blind person feel a vibration when there is an obstacle around them. These vibrators are connected to the Arduino, which is connected to the ultrasonic ranging modules. The ultrasonic ranging module works by providing accurate readings of a

distance. Therefore, it is programmed to send a pulse and if it detects an obstacle nearby it sends an alert to the vibrators. In addition, we used an Arduino to link the software and the hardware together. We also used a portable charger as a power supply and DC motor to keep the system working.

## 3.4 Project Overall Cost

The project was affordable but could have been costly if we used the haptic feedback instead of the vibrators. We ordered the vibrators online and took them two weeks to arrive so we did not have any issues with the delivery. As for the Arduino, we have it from before; we used it for other projects project. Furthermore, the mini DC motor is a cheap component to buy; it is an important component for our project. Furthermore, we also used an ultrasonic ranging module to detect obstacles, which is a big task in our project. It is a very cheap and useful component. Lastly, we use a portable charger that we had from before which we used daily to charge our phones or any other chargeable devices and it is expensive to buy. The table below (Table 3-6) shows the prices of each component used for the project and the overall cost.

*Table 3-6: Cost of components.*

| Components | Price |
|---|---|
| Mini Vibration Motor | 14 BD |
| Arduino Uno | 9 BD |
| Servo Motor SG 90 | 1 BD |
| Anker Portable Charger | 20 BD |
| Ultrasonic Ranging Module (X 2) | 1.9 * 2 = 3.8 BD |
| Total | 47.8 BD |

## 3.5 Summary

Overall, the components we chose relied on the cost, efficiency and accuracy. The Arduino was the most suitable choice as it is the link between the software and the hardware.

Furthermore, the Arduino controller was used to read from the coding program, produces the output, and sends it to the vibrators.

# 4 Chapter Four: System Implementation

## 4.1 Introduction

The ultrasonic sensors will be mounted on a vertical board which will be connected to the servo motor on the bottom to be able to move horizontally to the right and left and positioned in the front center of the vest. After that, the servo motor and ultrasonic sensors will be connected to the Arduino controller by wires. Then, the Arduino controller will be connected to the output delivery mechanism (i.e. the vibrators), which will be positioned on the inner side of the vest in a matrix-like shape. In addition, the Arduino controller will be programmed, connected to the battery and securely hidden in the vest. In figure 4-1, we can see a simple flowchart that demonstrates the Arduino program we used in this system. Also, figure 4-2 shows the block diagram for the components of our system. We can divide this system's functionality into three phases as follows:

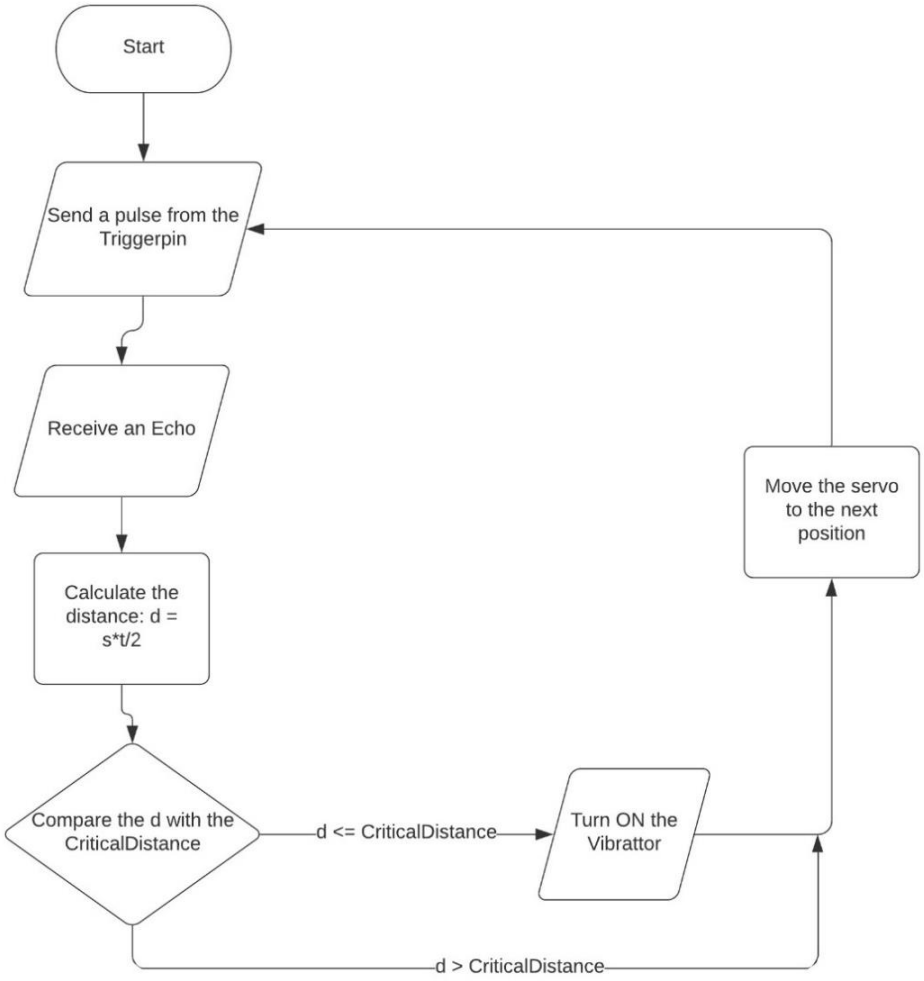


*Figure 4-1: system flowchart.*

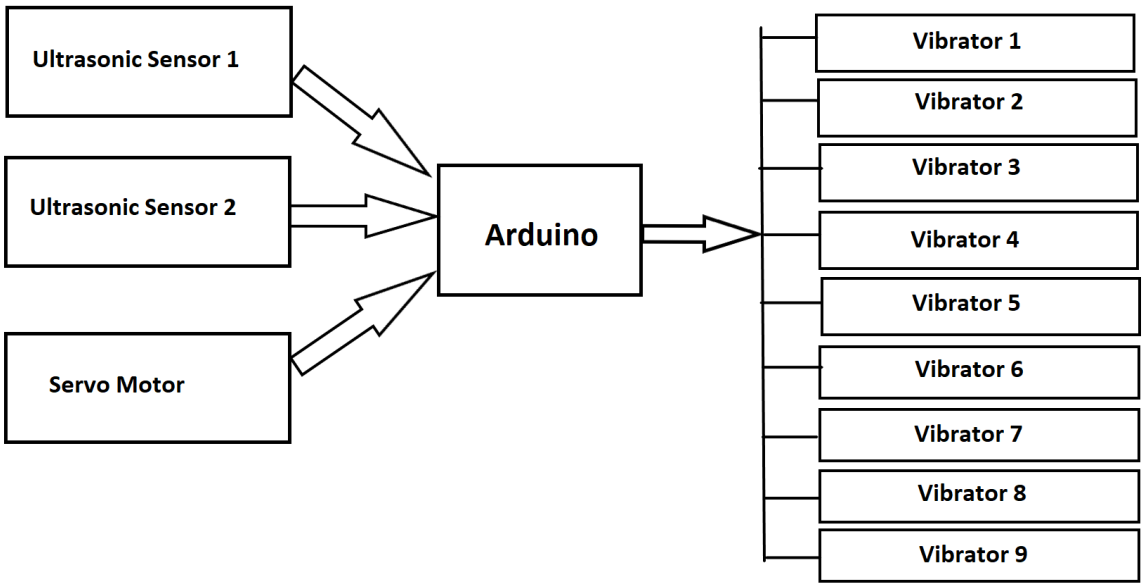


*Figure 4-2: Block Diagram.*

## 4.2 Phase one: collecting information (measurements of distances)

When the proposed system starts working in the first phase (figure 4-3), the sensors will be facing to the right side of the blind person. The first step in capturing the surrounding area will be measuring the distances of the obstacles to the right of the blind person using the two ultrasonic sensors which will yield three distances for three points, one in front of each sensor in addition to the point in the midpoint between the sensors. In the second step, the servo motor will move and change the angle of the sensors (i.e. to the front of the blind person) and capture the distances for three other points. Then, the second step will be repeated until the sensors are facing to the left. After that, the distances are stored in a matrix. Each element in that matrix represents a corresponding point in front of the blind person.

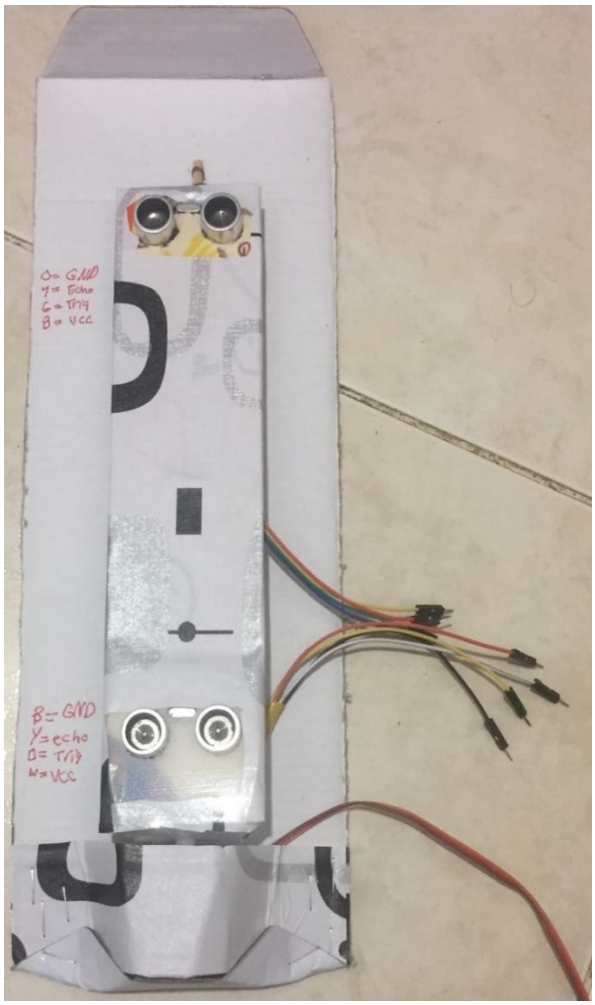


*Figure 4-3: the two ultrasonic ranger modules mounted on a piece of wood, which is connected to the servo motor.*

## 4.3 Phase Two: processing and encoding that information then calculating the outputs

In the second phase (figure 4-4), the Arduino controller will take that stored matrix of distances and compare each element in it with a threshold value (a safe distance for the blind person to be able to react in time) for example 1.5 meter (150 cm). If the measured distance is greater than the threshold value then it will be replaced with a zero value, and if it is smaller than the threshold value then it will be replaced with a one value.

*Figure 4-4: the vibrators connected to the Arduino.*

## 4.4 Phase Three: returning the outputs through the feedback vibrators

In the third phase (figure 4-5), the Arduino controller will only give feedback through the corresponding vibrator if the new value is one. In other words, the blind person will feel a vibration or an ultrasonic haptic feedback in the corresponding projected position of the obstacle in front of him, which should assist him in navigating his surroundings.

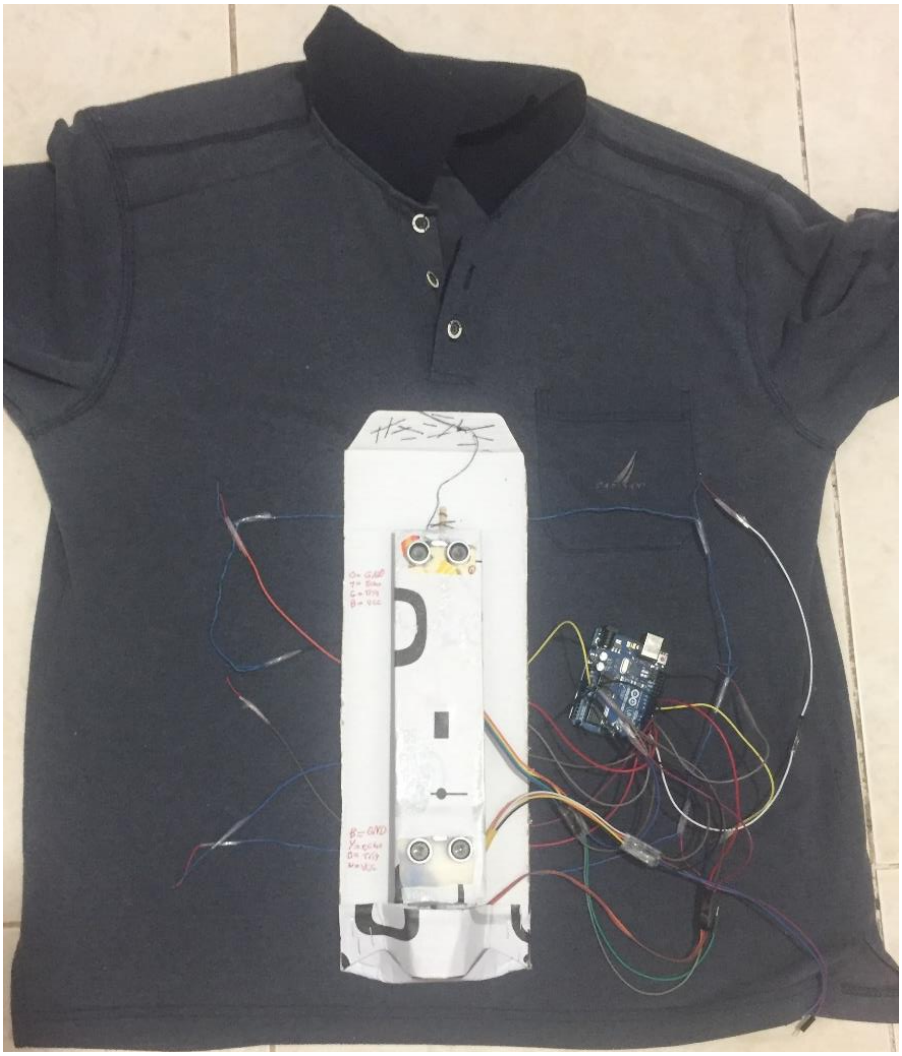

*Figure 4-5: the whole system connected (vibrators, the two ultrasonic ranger modules, and the servo motor all connected to the Arduino).*

## 4.5 Hardware connections

As we can see in figure 4-6 below, we have a matrix of points each point is attached to a vibrator from the inner side of the vest, which will give the feedback according to the distances from the obstacles in front of the blind person. We can also see the corresponding Arduino pins connected to the vibrators in table 4-1. In figure 4-7, we can see the two ultrasonic ranging modules connected to a board, which is intern fixed on the servo motor and mounted on the outer side of the vest. In addition, table 4-2 shows the specific wire connections to the Arduino that we used in this project.

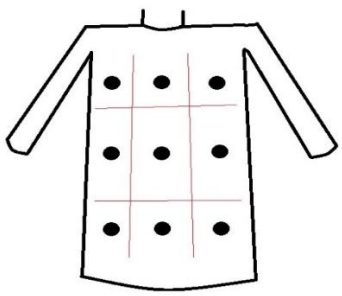

*Figure 4-6: the inner side of the vest (the vibrators) and the corresponding Arduino pin connected to it in the table.*

*Table 4-1: the order of Arduino pins connected to the vibrators.*

| | | |
|---|---|---|
| D4 | D5 | D6 |
| D7 | D8 | D9 |
| D10 | D11 | D12 |

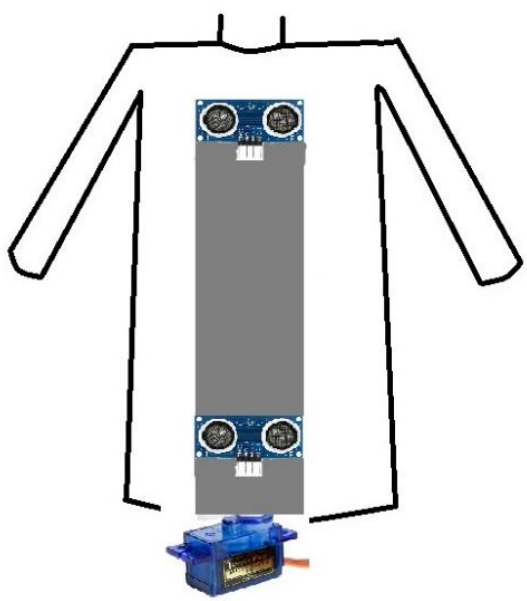

*Figure 4-7: the ultrasonic sensors and servo motor mounted on the outer side of the vest.*

*Table 4-2: The Arduino pins connections.*

| The Arduino pin | The wire it's connected to |
|---|---|
| D0 | Ultrasonic-1 Echo |
| D1 | Ultrasonic-1 Trig |
| D2 | Ultrasonic-2 Echo |
| D3 | Ultrasonic-2 Trig |
| D4 | Upper-level Right Vibrator VCC |
| D5 | Upper-level Middle Vibrator VCC |
| D6 | Upper-level Left Vibrator VCC |
| D7 | Mid-level Right Vibrator VCC |
| D8 | Mid-level Middle Vibrator VCC |
| D9 | Mid-level Left Vibrator VCC |
| D10 | Lower-level Right Vibrator VCC |
| D11 | Lower -level Middle Vibrator VCC |
| D12 | Lower -level Left Vibrator VCC |
| D13 | Servo Angle |

## 4.6 Results

After constructing the project prototype and programming the Arduino controller, we started using the prototype and running some tests with the help of some family members. We used a sleeping mask to cover the first test subject's eyes in order to simulate blindness while the other test subject was standing in some different positions around him simulating an obstacle. In conclusion, we found the following results:

The prototype functioned as we expected in alerting the first test subject to the position of the obstacle. As we can see in figures 4-8 to 4-11, the first test subject was able to correctly identify the position of the obstacle around him (the second test subject) and attempted to reach for it.

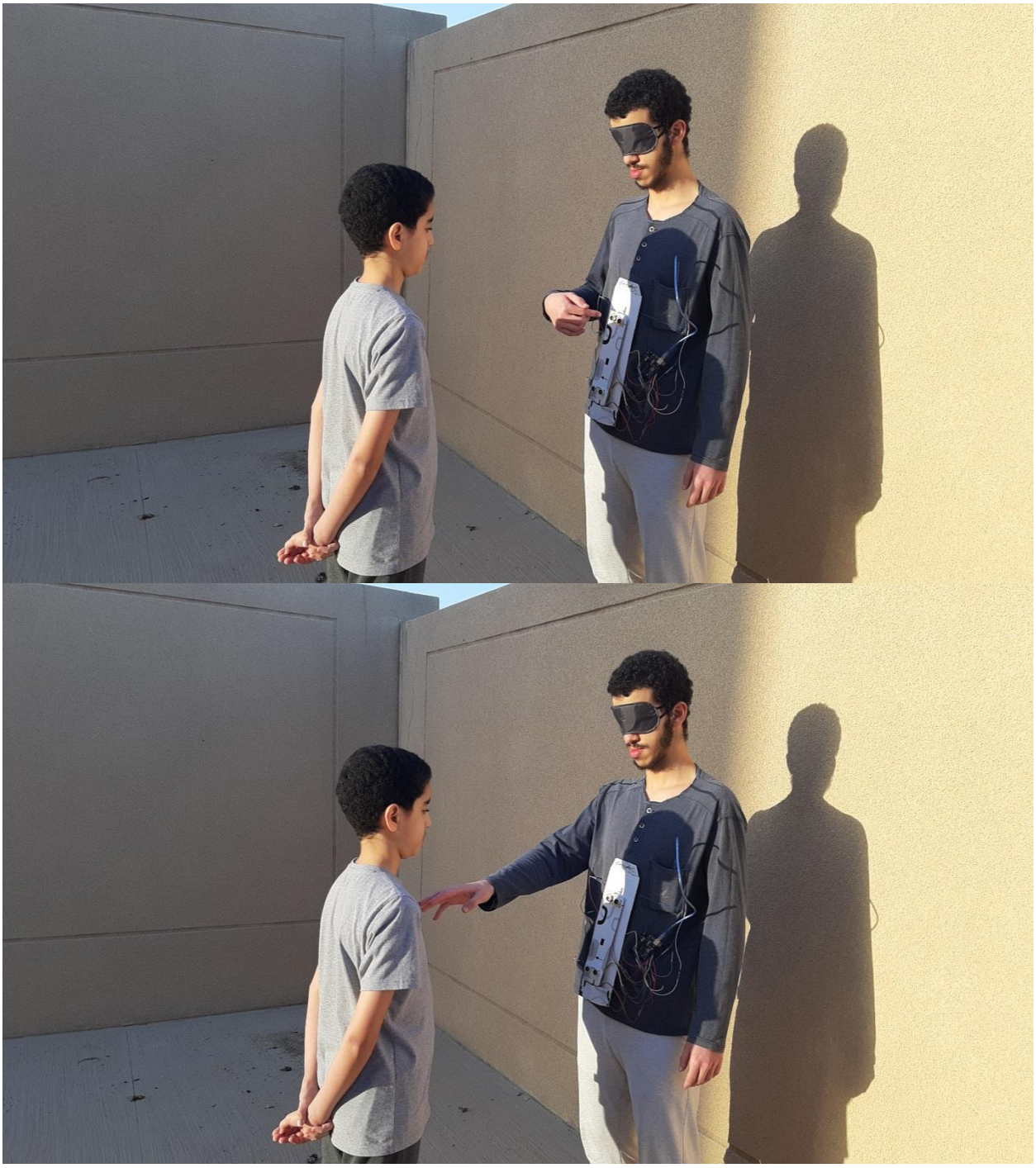

*Figure 4-8: the first test subject correctly identifies the position of the obstacle (in the center) and reaches for it.*

*Figure 4-9: the first test subject correctly identifies the position of the obstacle (to his left) and reaches for it.*

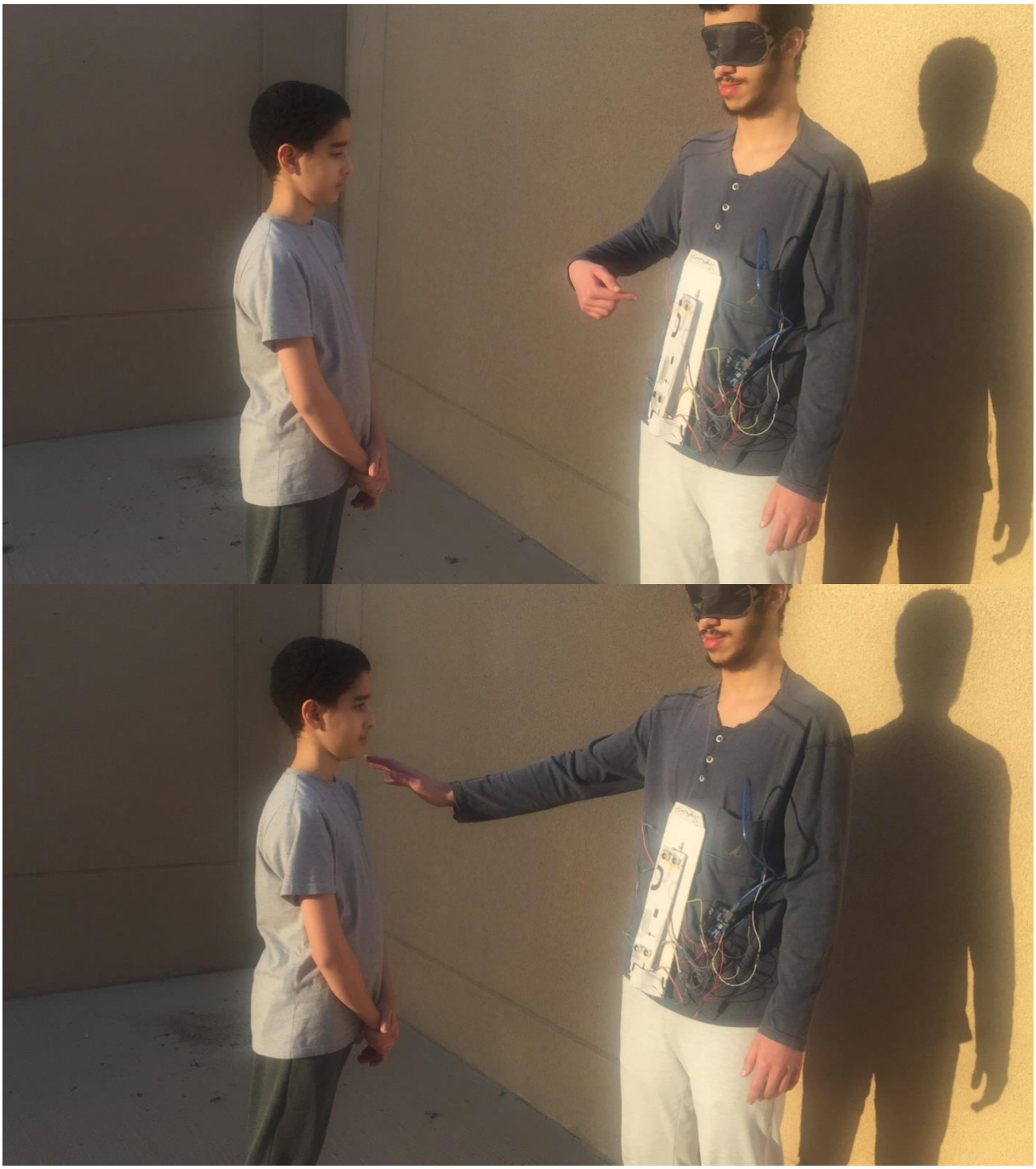

*Figure 4-10: the first test subject correctly identifies the position of the obstacle (to his right) and reaches for it.*

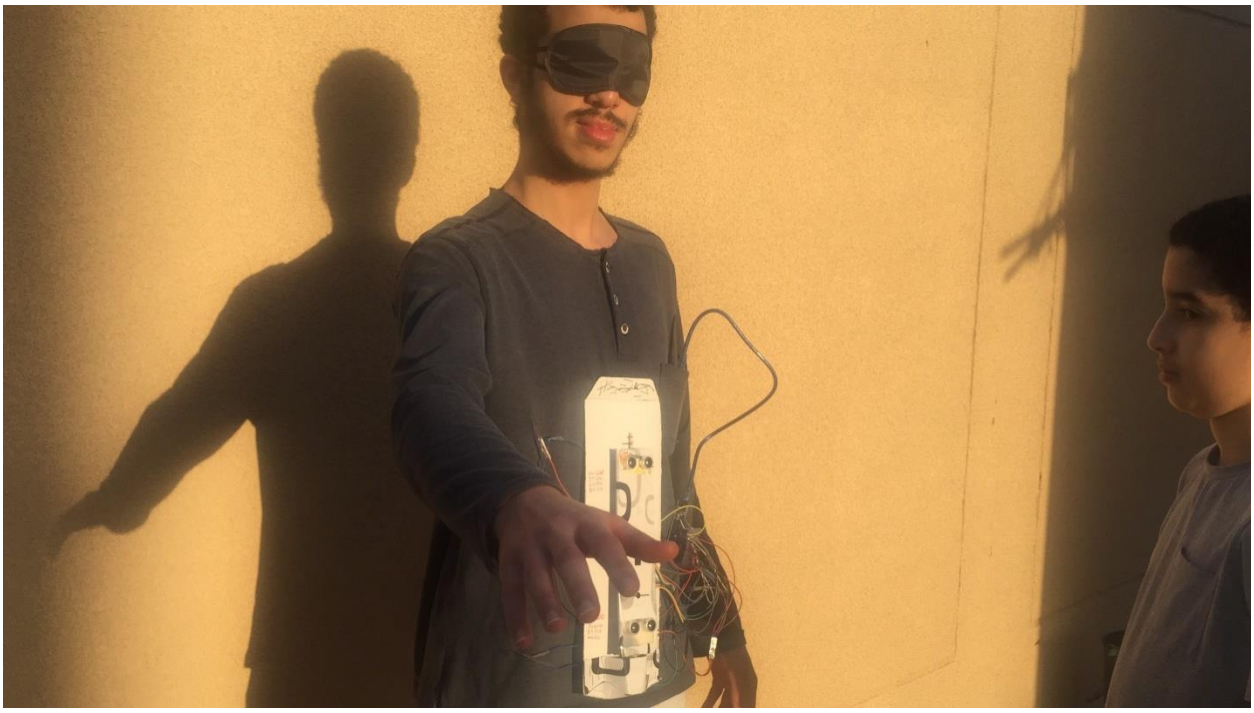

*Figure 4-11: the first test subject correctly identifies the position of the obstacle (to his right) and reaches for the camera operator.*

# 5 Chapter Five: Conclusion

## 5.1 Conclusion

In conclusion, the objective of this project is to try to help the disabled person to live a normal life. In addition, this project is just the beginning and got big potential in the future; it can be modified by using other technologies to help the disabled person to live happily and normally. The method that we used with the help of the mini vibrators, ultrasonic range module and other components is done as a concept that can be built on and can be upgraded financially by adding more vibrators or more advanced ultrasonic range module. Therefore, this shows that this system can be upgraded if we invest more on it to help the blind people.

## 5.2 Future work

This project is designed to be capable of assisting the blind and visually impaired people in navigating their surroundings. This system is developed by using vibrators to help feel if there was an obstacle around them. However, this system can be designed in many different methods. Future works can add more vibrators so that the person can be more accurate on where the obstacle is. Furthermore, another method to use is the haptic feedback instead of the vibrators. The haptic feedback will give you a better feel than the vibrators. Another method for the future work is to replace the ultrasonic sensor with Light Detection and Ranging (LIDAR). LIDAR measures obstacles with laser light and measures the reflection with a sensor. On the other hand, the ultrasonic sensor measures the distance by using ultrasonic waves. This shows that there are many methods of developing this system.

# Appendix

## The Arduino program used in this system

```
#include <Servo.h>

Servo myservo;  // create servo object to control a servo

#define echoPin1 0 // attach pin D0 Arduino to pin Echo of HC-SR04
#define trigPin1 1 //attach pin D1 Arduino to pin Trig of HC-SR04
#define echoPin2 2 // attach pin D2 Arduino to pin Echo of HC-SR04
#define trigPin2 3 //attach pin D3 Arduino to pin Trig of HC-SR04

// defines variables
long t1, t2, t3;            // variables for the duration of sound wave travel
int d1, d2, d3;             // variables for the distance measurement
int criticalDistance = 100;     // variables for the critical distance



void setup() {
  // Sets the vibrators as OUTPUTS
  pinMode(4,OUTPUT);
  pinMode(5,OUTPUT);
  pinMode(6,OUTPUT);
  pinMode(7,OUTPUT);
  pinMode(8,OUTPUT);
  pinMode(9,OUTPUT);
  pinMode(10,OUTPUT);
  pinMode(11,OUTPUT);
  pinMode(12,OUTPUT);
  myservo.attach(13);  // attaches the servo on pin 13 to the servo object
  pinMode(trigPin1, OUTPUT); // Sets the trigPins as OUTPUTS
  pinMode(trigPin2, OUTPUT);
  pinMode(echoPin1, INPUT); // Sets the echoPins as INPUTS
  pinMode(echoPin2, INPUT);

  // Serial Communication for testing
//  Serial.begin(9600);
//  Serial.println("Two Ultrasonic Sensors HC-SR04 Test");
//  Serial.println("with Arduino UNO R3");
}


void scan(){
```

```
//  ##########################
//  Scanning the Lower-level
//############################
  // Clears the trigPin condition
  digitalWrite(trigPin1, LOW);
  delayMicroseconds(10);
  // Sets the trigPin HIGH (ACTIVE) for 10 microseconds
  digitalWrite(trigPin1, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin1, LOW);
  // Reads the echoPin, returns the sound wave travel time in microseconds
  t1 = pulseIn(echoPin1, HIGH);
  // Calculating the distance
  d1 = t1 * 0.034 / 2; // Speed of sound wave (cm/µs) divided by 2 (go and back)
  // Displays the distance on the Serial Monitor
//  Serial.print("First radar Distance = ");
//  Serial.print(d1);
//  Serial.println(" cm");

//  ##########################
//  Scanning the Mid-level
//############################
   // Clears the trigPin condition
  digitalWrite(trigPin1, LOW);
  delayMicroseconds(10);
  // Sets the trigPin HIGH (ACTIVE) for 10 microseconds
  digitalWrite(trigPin1, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin1, LOW);
  // Reads the echoPin, returns the sound wave travel time in microseconds
  t3 = pulseIn(echoPin2, HIGH);
  // Calculating the distance
  d3 = t3 * 0.034 / 2; // Speed of sound wave (cm/µs) divided by 2 (go and back)
  // Displays the distance on the Serial Monitor
//  Serial.print("Mid-level Distance = ");
//  Serial.print(d3);
//  Serial.println(" cm");


//  ##########################
//  Scanning the Upper-level
//############################
   // Clears the trigPin condition
  digitalWrite(trigPin2, LOW);
  delayMicroseconds(10);
  // Sets the trigPin HIGH (ACTIVE) for 10 microseconds
```

```
  digitalWrite(trigPin2, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin2, LOW);
  // Reads the echoPin, returns the sound wave travel time in microseconds
  t2 = pulseIn(echoPin2, HIGH);
  // Calculating the distance
  d2 = t2 * 0.034 / 2; // Speed of sound wave (cm/µs) divided by 2 (go and back)
  // Displays the distance on the Serial Monitor
//  Serial.print("Second radar Distance = ");
//  Serial.print(d2);
//  Serial.println(" cm");
}

void loop() {
  myservo.write(160); //put the servo in the nutral position
  scan();            //start scaning fot distances
//  activating the feedback according to the criticalDistance
  if((d2 <= criticalDistance) && (d2 != 0)){
  digitalWrite(5,HIGH);}
  if((d3 <= criticalDistance) && (d3 != 0)){
  digitalWrite(8,HIGH);}
  if((d1 <= criticalDistance) && (d1 != 0)){
  digitalWrite(11,HIGH);}
  delay(1500);
  digitalWrite(5,LOW);
  digitalWrite(8,LOW);
  digitalWrite(11,LOW);


  myservo.write(240); //put the servo in the left position
  scan();            //start scaning fot distances
  //  activating the feedback according to the criticalDistance
  if((d2 <= criticalDistance) && (d2 != 0)){
  digitalWrite(6,HIGH);}
  if((d3 <= criticalDistance) && (d3 != 0)){
  digitalWrite(9,HIGH);}
  if((d1 <= criticalDistance) && (d1 != 0)){
  digitalWrite(12,HIGH);}
  delay(1500);
  digitalWrite(6,LOW);
  digitalWrite(9,LOW);
  digitalWrite(12,LOW);


  myservo.write(145);//put the servo in the right position
  scan();           //start scaning fot distances
```

```
  //  activating the feedback according to the criticalDistance
  if((d2 <= criticalDistance) && (d2 != 0)){
  digitalWrite(4,HIGH);}
  if((d3 <= criticalDistance) && (d3 != 0)){
  digitalWrite(7,HIGH);}
  if((d1 <= criticalDistance) && (d1 != 0)){
  digitalWrite(10,HIGH);}
  delay(1500);
  digitalWrite(4,LOW);
  digitalWrite(7,LOW);
  digitalWrite(10,LOW);
}
```

# References


1. Akash, S., & Rahman, M. (2018). QFD-Based Smart Cane Design: ATechnology to Assist Visually Impaired. *International Journal of BusinessAnalytics and Intelligence*, 6 (1), 25-30. http://publishingindia.com/ijbai/

2. Brito, D., Viana, T., Souza, D., Lourenco, A., & Paiva, S. (2018). A Mobile Solution To Help Visually Impaired People In Public Transports And Pedestrian Walks. *International Journal of Sustainable Development and Planning, 13*(2), 281-293

3. Dura-Gil, J.V., Bazuelo-Ruiz, B., Moro-Perez, D. & Molla-Domenech, F. (2017). Analysis of different vibration patterns to guide blind people. *PeerJ,*

4. Elmannai, W., & Elleithy, K. (2017). Sensor-based assistive devices for visually impaired people: Current status, challenges and future directions. *Sensors, 17(3),* 565.

5. Habib, A., Islam, M., Kabir, M., Mredul, M., & Al-Hasan, M. (2019). Staircase Detection to Guide Visually Impaired People: A Hybrid Approach. *Revue d'Intelligence Artificielle*, 33 (5), 327-334. https://doi.org/10.18280/ria.330501

6. Hoang, V., Nguyen, T., Le, T., Tran, T., Vuong, T., & Vuillerme, N.(2016). Obstacle detection and warning system for visually impairedpeople based on electrode matrix and mobile Kinect. *Vietnam J ComputSci*, 4, 71–83. DOI 10.1007/s40595-016-0075-z

7. Norberg, L., M.Sc, Westin, T., PhLic, Mozelius, P., M.Sc, & Wiklund, M., PhD. (2016). Web accessibility by morse code modulated haptics for deaf-blind. *International Journal of Child Health and Human Development*, 9(3), 361-369.

8. Petsiuk, A., & Pearce, J. (2019). Low-Cost Open Source Ultrasound-Sensing Based Navigational Support for the Visually Impaired. *Sensors*, 19 (17), 3783. doi:10.3390/s19173783

9. Ramadhan, A. (2018). Wearable Smart System for Visually Impaired People. *Sensors*, 18 (3), 843. doi:10.3390/s18030843

10. Tap, R., Mocanu, B., & Zaharia, T. (2017). A computer vision-based perception system for visually impaired. *Multimedia Tools and Applications* 76(9), 1171-11807

11. Quotefancy.com. 2020. Frederick Lenz Quote: "There Is A World Of Deaf, Dumb And Blind People Out There, Who Are Shuffling Through Their Lives, Not Knowing How Wonder...". [online] Available at: <https://quotefancy.com/quote/1749100/Frederick-Lenz-There-is-a-world-of-deaf-dumb-and-blind-people-out-there-who-are-shuffling> [Accessed 29 December 2020].

12. Who.int. 2020. Vision Impairment And Blindness. [online] Available at: <https://www.who.int/news-room/fact-sheets/detail/blindness-and-visual-impairment> [Accessed 29 December 2020].

13. Farnell.com. 2020. [online] Available at: <https://www.farnell.com/datasheets/1682209.pdf> [Accessed 29 December 2020].